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Data Science (BS)


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Affiliation: School of Science and Mathematics

MISSION OF THE STATISTICS & DATA SCIENCE PROGRAM

The mission of the Department of Computer and Data Sciences is to provide every student a foundation in data collection, visualization, modeling, analysis, and critical evaluation. Additionally, the department provides the foundation that students need to pursue graduate study and/or a career in statistics, biostatistics, data science, actuarial science, and other analytical fields.

THE MAJOR

Do you want to make meaning from complex data sets in computationally-rich fields? Data Science lies at the intersection of mathematics, statistics, computer science, and subject matter expertise. With a major in Data Science, you'll prepare for a career managing and processing large data sets, creating visualizations and models to identify patterns and trends, communicating the results to varied audiences, and making impactful decisions informed by data from diverse fields of knowledge, touching almost every aspect of the modern world.

Graduates should be well-qualified for advanced study in graduate programs or employment in a professional career requiring analytical skills. All data science graduates will obtain a background in mathematics, statistics, and computer science and gain experience with statistical computing, database management, and data manipulation and visualization. With a grounding in the liberal arts, the data science program also teaches critical thinking, problem solving, effective communication, and consideration of the ethical and moral implications of data science.

Truman's Data Science degree is consistent with the principles and recommendations contained in the American Statistical Association's "Curriculum Guidelines for Undergraduate Programs in Data Science," <https://www.amstat.org/asa/files/pdfs/EDU-DataScienceGuidelines.pdf>.

STATISTICAL COMMUNICATION and ANALYTIC STORYTELLING

All data science majors will develop their skills as speakers and as writers. DATA 344: Data Ethics is a writing-enhanced course where students develop their data communication skills. All students complete a capstone experience that includes a presentation and a paper. Many other courses in the major involve less formal writing and speaking experiences in a variety of settings, including longer and shorter forms, formal and informal styles, in-person and

online delivery, with a special emphasis on communication products aimed at both technical experts and non-experts.

ENRICHMENT OPPORTUNITIES

All Data Science students are encouraged to talk to their advisor about selecting a minor in a related area. Statistics uses tools from the mathematical sciences to examine and address problems in disciplines across campus. Students should also consider an internship, consulting, or other statistics and/or data science field experience, where they can apply their analytical know-how to real world situations.

On campus, our student-driven statistical consulting center, CASE, gives students a chance to work with clients from across campus with real data. Data Science contests such as ASA DataFest give students a chance to solve real world problems in a short time frame.

Students may obtain credit and research experience by participating in a recognized national undergraduate research program in data science, such as an REU site, or in one of several research programs sponsored by Truman.

There are also opportunities to work with faculty and other students as tutors or graders for lower-level statistics, computer science, and/or data science courses.

JOINT 4+1 GRADUATE/UNDERGRADUATE PROGRAMS

Truman offers a Master of Science in Data Science and Analytical Storytelling that prepares students for careers in a variety of industries in Data Science, Data Analytics, and related fields. Although this graduate program is open to students of any major, Data Science majors may use some courses to fulfill requirements for both programs. See the graduate section of catalog for more details on the MS in Data Science and Analytic Storytelling or talk to the department chair or one of our faculty members.

DEPARTMENTAL HONORS IN DATA SCIENCE

Honors in Data Science may be earned by:

1. Maintaining an overall grade point average of 3.5 or higher,
2. Maintaining a major grade point average of 3.5 or higher,
3. Demonstrating excellence in independent scholarship, and
4. Receiving the concurrence of the statistics and data science faculty.

DEGREE REQUIREMENTS:

Dialogues Requirements: 42-61 Credits

Missouri Statute: 0-4 credits

Bachelor of Science Requirement: 7 Credits

Courses listed in the Bachelor of Science Requirement may double count with minors or second majors, but may not double count with *Dialogues* requirements nor be used elsewhere in this major.

- **MATH 263 - Analytic Geometry and Calculus II** Credit(s): 4
OR **STAT 260 - Applied Mathematics for Data Analysis** Credit(s): 4
- **MATH 285 - Matrix Algebra** Credit(s): 3
OR **MATH 357 - Linear Algebra** Credit(s): 3

MAJOR REQUIREMENTS:

The Statistics major consists of three (3) parts: Required Support, Core Requirements, and Electives. Each student must complete all parts.

Part I: Required Support: 18-27 Credits

Courses listed in Required Support may double count for Dialogues, minors, and other campus requirements, but may not double count with requirements listed elsewhere in this major. Students should work with their advisor to pick the right set of courses for their future goals.

- **CS 180 - Foundations of Computer Science I** Credit(s): 4
- **MATH 198 - Analytic Geometry and Calculus I** Credit(s): 5
- **STAT 190 - Basic Statistics** Credit(s): 3
OR **STAT 290 - Statistics** Credit(s): 3

Choose and complete one or more of the following Specialization Blocks:

Intermediate Coding Block: 6 or more credit hours

- **CS 260 - Object-Oriented Programming and Design** Credit(s): 3
- **CS 310 - Data Structures and Algorithms** Credit(s): 3
OR **CS 480 - Artificial Intelligence** Credit(s): 3

Data Consulting Block: 6 or more credit hours

- **STAT 310 - Data Collection & Statistical Communication** Credit(s): 3
- **STAT 392 - Statistical Consulting with Practicum** Credit(s): 3

Statistics Block: 6 or more credit hours

Two of the following 400/500 level Statistics courses:

- **STAT 561 - Bayesian Statistics** Credit(s): 3
- **STAT 563 - Time Series Analysis with Markov Chains** Credit(s): 3
- **STAT 568 - Regression Analysis** Credit(s): 3
- **STAT 571 - Mathematical Probability and Statistics II** Credit(s): 3
- Another 400+ Statistics Course (with approval of the chair) Credit(s): 3

A Minor or Second Major: 15+ Credits

A minor or second major (15+ hours) in any field other than CS, Statistics, or Data Science. Minors in data-driven fields are particularly encouraged, including Astrophysics, Biology, Business, Economics, Health, Linguistics, Physics, Psychology, or Web Design UI/UX.

Learning Plan: 15+ Credits

A learning plan of 15 or more credit hours, approved by the advisor and chair, including at least six hours of STEM, broadly defined, and including at least one course at the 300+ level.

Note:

The computer science and statistics blocks are sufficient with coursework elsewhere in the major for students to concurrently earn minors in those areas.

Part II: Major Core Courses: 35 Credits

- **CS 181 - Foundations of Computer Science II** Credit(s): 3
- **CS 430 - Database Systems** Credit(s): 3
- **DATA 222 - Data Science** Credit(s): 3
- **DATA 322 - Intermediate Data Science** Credit(s): 3
- **DATA 324 - Data Visualization with Design** Credit(s): 3
- **DATA 344 - Data Ethics** Credit(s): 3
- **OR CS 345 - Cyberethics** Credit(s): 3
- **DATA 520 - Seminar: Data Mining and Multivariate Statistics** Credit(s): 3
- **STAT 101 - New Majors Seminar** Credit(s): 1
- **STAT 250 - Statistical Computing** Credit(s): 3
- **STAT 330 - Introduction to Linear Models** Credit(s): 3
- **OR STAT 331 - Biostatistics** Credit(s): 3
- **OR STAT 530 - Applied Statistical Methods I** Credit(s): 3
- **STAT 370 - Probability** Credit(s): 3
- **OR CS 191 - Computing Structures** Credit(s): 3
- **OR MATH 347 - Discrete Mathematics** Credit(s): 3
- **OR STAT 570 - Mathematical Probability and Statistics I** Credit(s): 3
- **STAT 398 - Seminar: Research, Reflection, and Career Readiness** Credit(s): 1
- **STAT 497 - Statistics Capstone Experience** Credit(s): 2
- **STAT 498 - Senior Seminar in Statistics and Data Science** Credit(s): 1

Part III: Major Electives: 6 Credits

- 6 additional credits from STAT, CS, MATH, or DATA not used elsewhere in this major, including at least 3 credits at the 300+ level.

Electives to Total: 120 Credits

Note: High-Impact Experiences and Capstones

Data Science students are encouraged to consider data-intensive High-Impact Experiences, such as internship and practicum experiences, research experiences, data contests (such as ASA Datafest), and consulting experiences (such as CASE).

Students who complete at least two credits of a data-intensive High-Impact Experience may request to have that experience substitute for **STAT 497**. Course credits may only be used once in the DATA major. All DATA students must complete **STAT 498**.

Those with a double-major may be able to design a single project that fulfils both majors's capstone expectation.

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